

LÊ BẢO PHÚC

Ho Chi Minh City, Vietnam | +84 366 856 827 | 24122021@student.hcmus.edu.vn
<https://github.com/BaoPhucx14022006> | <https://www.linkedin.com/in/bảo-phúc-lê-5972043b5/>

Education

University of Science - VNU HCM
Bachelor of Information Technology
GPA: 9.012/10.00

Ho Chi Minh City, Vietnam
Expected Graduation: 2028

Research Interests

Multimodal AI, Computer Vision, Natural Language Processing, Generative AI, Large Language Models.

Technical Skills

Programming: Python, C++, SQL

AI / Machine Learning: PyTorch, Transformers, Computer Vision, NLP, LLMs, Embeddings, RAG

AI Frameworks / Models: Langchain, FAISS, Hugging Face Transformers

Backend / Databases: FastAPI, SQLite, NoSQL

Tools: Git, GitHub

Projects

AIC Multimodal Video Retrieval System

2026

- Preprocessed video datasets by sampling keyframes, transcribing Vietnamese speech with Whisper, and extracting on-screen text using OCR.
- Extracted visual embeddings with CLIP and SigLIP2, and built text indices using BM25 and RapidFuzz for fuzzy keyword matching.
- Designed a KIS (Known-Item Search) pipeline that splits chronological queries into sequential events and filters results by time order.
- Combined multimodal retrieval results using Reciprocal Rank Fusion (RRF) to pinpoint the exact target video segments quickly.

ViFinQA Text-to-Pandas System

2026

- Built a 2-stage Text-to-Pandas pipeline that answers Vietnamese financial questions (ViFinQA) by converting natural language into executable pandas code over extracted financial-statement tables.
- Designed an LLM-based query parser (Qwen) that extracts tickers, years, report scope, and target metrics from Vietnamese questions, backed by a rule-based fallback for guaranteed entity extraction.
- Implemented a document/table indexer and retriever that grounds each query in the correct financial statement, ranks candidate tables by keyword relevance, and exports evidence as CSV.
- Generated and safely executed pandas code (DeepSeek) with a sandboxed evaluator, unit normalization, and a heuristic fallback engine to guarantee a numeric answer for every query.
- Built a flexible LLM client supporting OpenAI-compatible, Ollama, and direct HuggingFace GPU inference (Kaggle/Colab), and a benchmark evaluator with automated submission packaging.

Document Summarization & RAG Chatbot

2026

- Built a document summarization and Q&A chatbot that ingests PDF/DOCX files and generates concise Vietnamese summaries using NVIDIA-hosted LLMs (Llama 3.1 70B) via LangChain.
- Implemented semantic chunking of extracted text using embedding-based breakpoint detection to preserve context across long documents before summarization.
- Designed a hierarchical map-reduce summarization strategy, summarizing each chunk individually then synthesizing a final summary across all chunks.
- Built a RAG retrieval pipeline with Qdrant vector store and NVIDIA embeddings, using similarity-score-thresholded retrieval to answer follow-up questions grounded in the document.
- Persisted multi-turn chat history using SQLite and structured prompt templates (YAML) for summarization and retrieval tasks.